Practical 4.1: CRD as a WMS Basemap — Leaflet Web App

Build a Leaflet map with Mwanza and Tanga layers from CRD

Day 4 · 3 hours · Resilience Academy SDI Training

Reference: `crd.resilienceacademy.ac.tz` | Data: Tanga & Mwanza, Tanzania

Learning Objectives:

- Understand how WMS tile services work as basemaps
- Build a Leaflet web application using CRD WMS layers
- Load Mwanza business and education layers from the live CRD
- Add Tanga satellite imagery and implement a layer toggle control

WMS Basemap Concept

A **basemap** is a background reference layer. The CRD exposes every uploaded layer as a WMS endpoint automatically. Any application supporting WMS — Leaflet, QGIS, ArcGIS, Google Maps API — can use your CRD as a data source.

```
# Test a WMS endpoint in your browser or with curl:
curl -s "https://crd.resilienceacademy.ac.tz/geoserver/ows?\
SERVICE=WMS&VERSION=1.1.1&REQUEST=GetCapabilities" | \
grep -o 'geonode:[^"<]*' | head -20
```

Building the Leaflet Web App

Create `/sdi-training/mwanza-tanga-map.html`:

```
<!DOCTYPE html>
<html>
<head>
  <title>Mwanza & Tanga --- CRD Basemap</title>
  <meta charset="utf-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <link rel="stylesheet"
    href="https://unpkg.com/leaflet@1.9.4/dist/leaflet.css"/>
  <script src="https://unpkg.com/leaflet@1.9.4/dist/leaflet.js"></script>
  <style>
    body { margin: 0; padding: 0; }
    #map { position: absolute; top: 0; bottom: 0; width: 100%; }
    #info { position: absolute; top: 10px; left: 50px; z-index: 1000;
      background: white; padding: 10px; border-radius: 5px;
      border: 2px solid #007064; font-family: Arial; }
  </style>
</head>
<body>
<div id="info">
  <b style="color:#007064">Resilience Academy CRD</b><br>
  <small>Tanga & Mwanza Data</small>
</div>
<div id="map"></div>
<script>
```

```
// Initialize map centred between Tanga and Mwanza
var map = L.map('map').setView([-4.5, 35.5], 6);

// Background: OpenStreetMap
var osm = L.tileLayer(
  'https://{s}.tile.openstreetmap.org/{z}/{x}/{y}.png',
  {attribution: '&copy; OpenStreetMap contributors', maxZoom: 19}
).addTo(map);

var CRD = 'https://crd.resilienceacademy.ac.tz/geoserver/ows';

// Mwanza Businesses layer
var mwanzaBiz = L.tileLayer.wms(CRD, {
  layers: 'geonode:mwanza_businesses',
  format: 'image/png',
  transparent: true,
  opacity: 0.85,
  attribution: 'Resilience Academy CRD'
});

// Mwanza Education Facilities
var mwanzaEdu = L.tileLayer.wms(CRD, {
  layers: 'geonode:mwanza_education_facilities',
  format: 'image/png',
  transparent: true,
  opacity: 0.85,
  attribution: 'Resilience Academy CRD'
});

// Tanga Satellite Imagery
var tangaImg = L.tileLayer.wms(CRD, {
  layers: 'geonode:Tanga_imagery_reduced',
  format: 'image/png',
  transparent: true,
  opacity: 0.9,
  attribution: 'Resilience Academy / KOMPSAT-3'
});

// Layer control
var basemaps = { "OpenStreetMap": osm };
var overlays = {
  "Mwanza Businesses": mwanzaBiz,
  "Mwanza Education Facilities": mwanzaEdu,
  "Tanga Satellite Imagery": tangaImg
};
L.control.layers(basemaps, overlays).addTo(map);

// Add Mwanza businesses by default
mwanzaBiz.addTo(map);

// GetFeatureInfo popup on click
map.on('click', function(e) {
  var url = CRD + '?SERVICE=WMS&VERSION=1.1.1&REQUEST=GetFeatureInfo' +
    '&LAYERS=geonode:mwanza_businesses' +
    '&QUERY_LAYERS=geonode:mwanza_businesses' +
    '&INFO_FORMAT=text/plain' +
    '&X=50&Y=50&WIDTH=101&HEIGHT=101' +
    '&BBOX=' + (e.latlng.lng-0.01)+','+(e.latlng.lat-0.01)+','
           + (e.latlng.lng+0.01)+','+(e.latlng.lat+0.01) +
```

```
    '&SRS=EPSG:4326';
  fetch(url)
    .then(r => r.text())
    .then(text => {
      if (text && text.includes('=')) {
        L.popup().setLatLng(e.latlng)
          .setContent('<pre style="max-width:300px;font-size:11px;">'
            + text.substring(0, 500) + '</pre>')
          .openOn(map);
      }
    });
});
</script>
</body>
</html>
```

```
# Open in browser (Linux):
xdg-open ~/sdi-training/mwanza-tanga-map.html
# Or from Windows WSL2:
# Copy to Windows and open with Chrome/Firefox
```

Checkpoint:

- ✓ HTML map file opens in browser
- ✓ Mwanza businesses layer visible as coloured points on map
- ✓ Mwanza education facilities layer can be toggled on/off
- ✓ Tanga satellite imagery loads when toggled on
- ✓ Clicking near a Mwanza feature shows attribute popup